

A five-dimensional counterexample to greedy suboptimality for sparse approximation triples

Solution packet for arXiv:1506.03549, Appendix B

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Result. The greedy estimate (B.4) in arXiv:1506.03549 is false for general sparse approximation triples, even in dimension five and even when every signal is compressible. All greedy choices in the example are unique. For a family x_ε , the two-step greedy/best-error ratio is $(1 + \varepsilon)/\varepsilon \rightarrow \infty$; at three steps the best error is zero but the greedy error remains positive.

1. The question

Sun and Tang define, for a sparse approximation triple $(\mathbf{A}, \mathbf{M}, \mathbf{H})$, the greedy iterates

$$x^{k+1} = x^k + \operatorname{argmin}_{a \in \mathbf{A}} \|x - x^k - a\|_{\mathbf{M}}, \quad x^0 = 0.$$

Their Appendix B asks whether there is a constant $C > 0$ such that

$$\|x^k - x\|_{\mathbf{M}} \leq C \sigma_{k\mathbf{A}, \mathbf{M}}(x), \quad \sigma_{k\mathbf{A}, \mathbf{M}}(x) := \inf_{z \in k\mathbf{A}} \|x - z\|_{\mathbf{M}}, \quad (1)$$

even for compressible x [1]. Here is the exact question from page 27 of the official arXiv PDF.

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Given a sparse approximation triple $(\mathbf{A}, \mathbf{M}, \mathbf{H}_1)$, we say that $x \in \mathbf{M}$ is *compressible* ([8] [25] [26] [42]) if $\{\sigma_{k\mathbf{A}, \mathbf{M}}(x)\}_{k=1}^\infty$ having rapid decay, such as

$$\sigma_{k\mathbf{A}, \mathbf{M}}(x) \leq Ck^{-\alpha} \text{ for some } C, \alpha > 0,$$

where $\sigma_{k\mathbf{A}, \mathbf{M}}(x)$ is the best approximation error of x from $k\mathbf{A}$,

$$(B.2) \quad \sigma_{k\mathbf{A}, \mathbf{M}}(x) := \inf_{\hat{x} \in k\mathbf{A}} \|\hat{x} - x\|_{\mathbf{M}}, \quad k \geq 1.$$

For the sequence $x_{\mathbf{A}, \mathbf{M}}^k, k \geq 0$, in the greedy algorithm ([5.3]), we have

$$(B.3) \quad \|x_{\mathbf{A}, \mathbf{M}}^k - x\|_{\mathbf{M}} \geq \sigma_{k\mathbf{A}, \mathbf{M}}(x),$$

as $x_{\mathbf{A}, \mathbf{M}}^k \in k\mathbf{A}, k \geq 1$. The above inequality becomes an equality in the classical sparse recovery setting. We do not know whether and when the greedy algorithm ([5.3]) is suboptimal, i.e., there exists a positive constant C such that

$$(B.4) \quad \|x_{\mathbf{A}, \mathbf{M}}^k - x\|_{\mathbf{M}} \leq C \sigma_{k\mathbf{A}, \mathbf{M}}(x), \quad x \in M,$$

even for compressible signals. The reader may refer to [54] for the study of various greedy algorithms.

Official source evidence: arXiv:1506.03549, p. 27.

2. Proof intuition

Coordinate subspaces turn this greedy algorithm into weighted greedy set cover. In ℓ_1 , the best approximation on a coordinate support simply keeps the coordinates on that support. Thus a greedy step chooses the allowed support containing the largest remaining ℓ_1 mass, whereas a best k -term approximation chooses k supports whose *union* captures the most mass. Overlapping supports can lure greedy into covering two large coordinates that the optimal pair would cover separately.

The only delicate issue is whether such a support family satisfies every axiom imposed on a sparse approximation triple. It does in finite dimension: coordinate projections are simultaneously best in ℓ_1 and ℓ_2 , both required norm-splitting identities are exact, and all closed sets are proximal.

3. The fixed triple

Let

$$\mathbf{M} = (\mathbb{R}^5, \|\cdot\|_1), \quad \mathbf{H} = (\mathbb{R}^5, \|\cdot\|_2).$$

For $S \subset \{1, \dots, 5\}$, write $E_S = \{z \in \mathbb{R}^5 : \text{supp } z \subset S\}$. Define

$$S_1 = \{1, 2\}, \quad S_2 = \{3, 4\}, \quad S_3 = \{1, 3\}, \quad S_4 = \{5\},$$

and $\mathbf{A} = E_{S_1} \cup E_{S_2} \cup E_{S_3} \cup E_{S_4}$.

Lemma 1. $(\mathbf{A}, \mathbf{M}, \mathbf{H})$ is a sparse approximation triple in the sense of the source. Moreover $3\mathbf{A} = \mathbb{R}^5$.

Proof. We verify the five source axioms.

- (i) The inclusions $\mathbf{A} \subset \mathbf{M} \subset \mathbf{H}$ are continuous: $\|z\|_2 \leq \|z\|_1$, and on every E_{S_i} , $\|z\|_1 \leq \sqrt{2}\|z\|_2$.
- (ii) $\mathbf{A}$ is a finite union of closed subspaces. Every nonempty closed subset of finite-dimensional $\mathbf{M}$ is proximal: a minimizing sequence for distance from a fixed point is bounded and has a convergent subsequence.
- (iii) If P_S denotes coordinate projection, then $P_S x$ is the unique best approximation to x from E_S in both ℓ_1 and ℓ_2 .
- (iv) Coordinate separation gives exactly

$$\|x\|_1 = \|P_S x\|_1 + \|x - P_S x\|_1, \quad \|x\|_2^2 = \|P_S x\|_2^2 + \|x - P_S x\|_2^2.$$

- (v) Since $S_1 \cup S_2 \cup S_4 = \{1, \dots, 5\}$, $E_{S_1} + E_{S_2} + E_{S_4} = \mathbb{R}^5$. Hence $3\mathbf{A} = \mathbf{H}$, which is stronger than the required density of $\bigcup_k k\mathbf{A}$.

□

For any residual r and allowed support S_i ,

$$\inf_{a \in E_{S_i}} \|r - a\|_1 = \|r - P_{S_i} r\|_1 = \|r\|_1 - \sum_{j \in S_i} |r_j|. \quad (2)$$

Thus the greedy support is exactly the one capturing the greatest remaining coordinate mass.

4. The counterexample

Fix $0 < \varepsilon < 1$ and put

$$x_\varepsilon = (2, \tfrac{3}{2}, 2, 1, \varepsilon).$$

The captured masses at the first three greedy steps are:

k	residual	S_1	S_2	S_3	S_4	unique choice
0	$(2, \frac{3}{2}, 2, 1, \varepsilon)$	$\frac{7}{2}$	3	4	ε	S_3
1	$(0, \frac{3}{2}, 0, 1, \varepsilon)$	$\frac{3}{2}$	1	0	ε	S_1
2	$(0, 0, 0, 1, \varepsilon)$	0	1	0	ε	S_2

The inequalities are strict because $0 < \varepsilon < 1$. Hence the two- and three-step greedy residuals are

$$r^2 = (0, 0, 0, 1, \varepsilon), \quad r^3 = \varepsilon e_5, \quad (3)$$

so their errors are $1 + \varepsilon$ and ε .

Theorem 1. *For the fixed sparse approximation triple of Lemma 1, no constant C can satisfy (1), even when restricted to compressible signals and even under unique greedy choices.*

Proof. The pair S_1, S_2 covers coordinates 1, 2, 3, 4, so $\sigma_{2\mathbf{A}, \mathbf{M}}(x_\varepsilon) \leq \varepsilon$. Every other union of two allowed supports misses at least one of those four coordinates, whose weight is at least one. Therefore

$$\sigma_{2\mathbf{A}, \mathbf{M}}(x_\varepsilon) = \varepsilon, \quad \text{quad} \frac{\|x^2 - x_\varepsilon\|_1}{\sigma_{2\mathbf{A}, \mathbf{M}}(x_\varepsilon)} = \frac{1 + \varepsilon}{\varepsilon} \rightarrow \infty. \quad (4)$$

This already rules out a uniform C while the denominator is positive.

There is also a pointwise exact-sparsity failure. Because $x_\varepsilon \in E_{S_1} + E_{S_2} + E_{S_4} \subset 3\mathbf{A}$,

$$\sigma_{3\mathbf{A}, \mathbf{M}}(x_\varepsilon) = 0, \quad \|x^3 - x_\varepsilon\|_1 = \varepsilon > 0 \quad (5)$$

by (3). Hence no finite constant works even for a fixed x_ε .

Finally, every $x \in \mathbf{M}$ is compressible in the source's sense. Indeed, $3\mathbf{A} = \mathbf{M}$ implies $\sigma_{k\mathbf{A}, \mathbf{M}}(x) = 0$ for $k \geq 3$; for any $\alpha > 0$, a finite enlargement of the constant handles $k = 1, 2$ in $\sigma_{k\mathbf{A}, \mathbf{M}}(x) \leq C_x k^{-\alpha}$. Thus (4)–(5) occur entirely inside the requested compressible class. $\square$

5. Structural meaning and reproducibility

More generally, any finite family of coordinate supports whose finite unions span $\mathbb{R}^d$ gives a sparse approximation triple with $\mathbf{M} = \ell_1^d$, $\mathbf{H} = \ell_2^d$. Formula (2) identifies the source algorithm with weighted greedy maximum coverage, while best approximation from $k\mathbf{A}$ is optimal weighted coverage by k allowed supports. Consequently a positive theorem needs an additional structural hypothesis on the support family; the five axioms alone cannot imply (1).

The included exact-rational script `code/verify_counterexample.py` enumerates all support unions, confirms the three strict greedy choices, and verifies (4)–(5) for several rational ε . Run:

```
python3 code/verify_counterexample.py
```

The expected final line is `coordinate-group greedy counterexample: PASS`. The script is an audit aid; the proof above is exact and self-contained.

Exact-id and exact-pharse searches of the run indexes found no duplicate. A bounded public search through 2026-08-13 found the arXiv and published source but no later answer to this exact sparse-approximation-triple question. The claim here is the negative result above, not a characterization of support families for which a positive Lebesgue bound holds.

References

- [1] Q. Sun and W.-S. Tang, *Nonlinear frames and sparse reconstructions in Banach spaces*, arXiv:1506.03549; J. Fourier Anal. Appl. 23 (2017), 1118–1152, especially (5.3) and Appendix B, <https://arxiv.org/abs/1506.03549>.